DERIVED COMPLETENESS WITH RESPECT TO I EXTENDED TO B IS DETECTED BY THE RESTRICTION FUNCTOR

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A complex L over B lies in D_comp(B, IB) if and only if T(L, f) vanishes for every local section f of the ideal IB generated by I. invoked at proof lines 26623-26626. Reduces membership to the vanishing of explicit functors.

Source reference. Original lemma 091Q: sources/stacks-more-algebra/pinned-original-source.tex, source lines 25936–25936. Cited at proof lines 26623-26624 for the vanishing characterization of membership; the conclusion is used inside the counted span.

Lemma 0.2. *Let $A \to B$ be a ring map. Let $I \subset A$ be an ideal. The inverse image of $D_{comp}(A, I)$ under the restriction functor $D(B) \to D(A)$ is $D_{comp}(B, IB)$.*

Proof. Using Lemma 091Q we see that $L \in D(B)$ is in $D_{comp}(B, IB)$ if and only if $T(L, f)$ is zero for every local section $f \in I$. Observe that the cohomology of $T(L, f)$ is computed in the category of abelian groups, so it doesn't matter whether we think of f as an element of A or take the image of f in B. The lemma follows immediately from this and the definition of derived complete objects. □

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